Q1.

An operating system schedules processes so that the processor is never idle while runnable work exists. The scheduler chooses which ready process runs next.

Preemptive scheduling allows the operating system to interrupt a running process, whereas non preemptive scheduling waits for it to yield voluntarily.

Q2.

Compare the scheduling algorithms in the table below.

Algorithm	Preempt	Avg Wait
FCFS	No	8.0
SJF	No	6.2
RR	Yes	7.4

a)

Shortest job first gives the lowest average waiting time of the three.

It does however require knowing the burst time of each process in advance, which a real operating system cannot generally do, so it is approximated.

b)

Round robin is preferred for interactive systems because every process receives the processor within a bounded time.

Rough work: quantum = 4 ms.

Q3.

The process state pipeline is shown below.